WEEK 05 · SESSION 2

Regularized and Nonlinear Regression

Estimate continuous outcomes with interpretable baselines and robust evaluation.

DSACADEMY

Learning outcomes

01 Explain the bias-variance tradeoff.

02 Use regularization to stabilize coefficients.

03 Compare tree regressors with linear models.

The concept system

01

regularization

02

bias and variance

03

trees

04

cross-validation

FROM QUESTION TO EVIDENCE

A repeatable reasoning loop

01 **Define** → 02 **Transform** → 03 **Validate** → 04 **Communicate**

Compare ridge, lasso, and gradient boosting on house prices.

Guided lab

LAB BRIEF

Compare ridge, lasso, and gradient boosting on house prices.

workbook / tests / evidence

01 **Frame**

Define inputs, outputs, and success criteria.

02 **Build**

Implement the smallest correct workflow.

03 **Verify**

Test normal, boundary, and failure cases.

04 **Explain**

Record the evidence and remaining limits.

Portfolio deliverable

A model comparison table and justified selection.

QUALITY GATE

- Reproducible
- Tested
- Decision-relevant

WEEKLY ASSESSMENT

Regression Model Card

Evaluation 30%, diagnostics 25%, model choice 20%, reproducibility 15%, communication 10%.

SESSION CHECKPOINT

What should remain after the lesson?

CONCEPT	regularization + bias and variance
PRACTICE	Compare ridge, lasso, and gradient boosting on house prices.
EVIDENCE	A model comparison table and justified selection.